

Supporting Information-1

Aggregation Rules of Short Peptides

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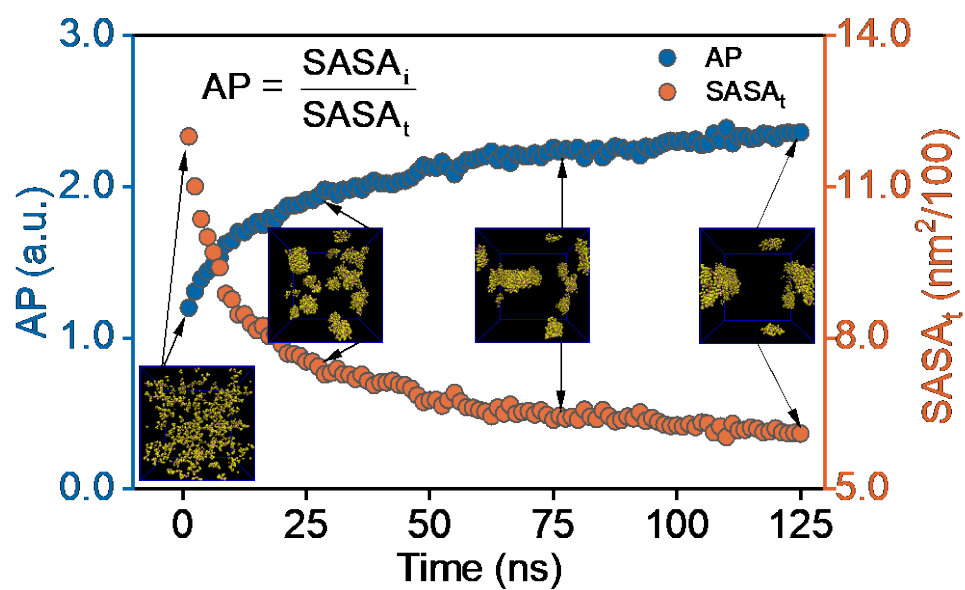


FIGURE S1 Example of an aggregating peptide illustrating the evolution of aggregation propensity (AP) and accessible surface area (SASA). The AP is calculated as the ratio between the initial accessible surface area (SASA_i) and accessible surface area at time t (SASA_t) during a CGMD simulation.

实验只是0或1；模拟没考虑浓度不一样？
30mM就聚了的，模拟基本在2以上，说明确实聚集
但确实数据集中都是些强疏水的，像纯Q纯R这种模拟做不出来

TABLE S1 Comparison of 165 predicted and experimental results of peptide aggregation. The 165 experimental results are obtained from our previous work¹. The 0 and 1 in “Aggre-exp” and “Aggre-sim” columns indicate non-aggregation and aggregation (if $AP_{\text{prd}} > 1.4$), respectively. The last column “Compare” indicates the consistency of the predicted and experimental results, *i.e.*, if they are consistent, the “Compare” value will be 1, otherwise it is 0. Out of the 165 peptides, only 10 cases are not consistent, yielding an accuracy percentage of ~94%.

Index	Sequence	Concentration (M)	AP_{prd}	Aggre-exp	Aggre-sim	Compare
1	SSSS	120	1.8332	0	1	0
2	DDDD	120	0.9319	1	0	0
3	EEEE	120	0.8851	1	0	0
4	GGGG	120	1.3248	1	0	0
5	NNNN	120	1.2591	1	0	0
6	QQQQ	120	1.0772	1	0	0
7	RRRR	120	1.0913	1	0	0
8	DALV	120	1.275	1	0	0
9	MVQG	120	1.3525	1	0	0
10	VALE	120	1.2996	1	0	0
11	AAAA	120	1.005	0	0	1
12	KKKK	120	1.0727	0	0	1
13	EEKK	120	1.2537	0	0	1
14	PWWY	120	2.3536	1	1	1
15	LWFW	30	2.3645	1	1	1
16	WVFY	30	2.3512	1	1	1
17	PVFY	30	2.3566	1	1	1
18	WAYF	30	2.2872	1	1	1
19	WEFF	30	2.1606	1	1	1
20	FKFF	60	2.2156	1	1	1
21	FKFW	60	2.2171	1	1	1
22	WKFF	60	2.2172	1	1	1
23	YAWF	60	2.2578	1	1	1
24	CCCC	120	1.9157	1	1	1
25	HHHH	120	1.9015	1	1	1
26	PPPP	120	1.8221	1	1	1
27	TTTT	120	1.8106	1	1	1
28	WWWW	120	2.2982	1	1	1
29	FFFF	30	2.3793	1	1	1
30	IIII	30	1.9876	1	1	1
31	LLLL	30	1.9	1	1	1
32	VVVV	60	2.0349	1	1	1
33	MMMM	120	1.7721	1	1	1
34	YYYY	120	2.274	1	1	1
35	CWFT	120	2.2305	1	1	1
36	WFHP	120	2.2258	1	1	1
37	FGWW	120	2.256	1	1	1
38	IRVW	120	2.0482	1	1	1

39	CHFW	30	2.2302	1	1	1
40	DVFW	30	2.1214	1	1	1
41	NWWF	30	2.2206	1	1	1
42	LSFW	60	2.2793	1	1	1
43	LFWS	120	2.2579	1	1	1
44	CWHY	120	2.1951	1	1	1
45	WALW	120	2.22	1	1	1
46	EFFK	120	2.0727	1	1	1
47	FFYY	30	2.3623	1	1	1
48	WYFF	30	2.3707	1	1	1
49	LVFF	30	2.3513	1	1	1
50	GFFY	30	2.2446	1	1	1
51	LGVF	120	2.1237	1	1	1
52	NVFH	120	2.0695	1	1	1
53	RWVF	120	2.0954	1	1	1
54	TAVF	120	2.0911	1	1	1
55	VFHN	120	2.0761	1	1	1
56	VFMM	120	2.1607	1	1	1
57	YVFA	120	2.1881	1	1	1
58	LVFG	60	2.1261	1	1	1
59	MVFP	60	2.2009	1	1	1
60	FMLF	30	2.284	1	1	1
61	LLLS	30	1.9345	1	1	1
62	IFYT	60	2.2499	1	1	1
63	LLSL	60	1.9334	1	1	1
64	VWIT	120	2.1667	1	1	1
65	WLVW	90	2.2993	1	1	1
66	LWWF	30	2.3586	1	1	1
67	VFVF	30	2.3151	1	1	1
68	MMVV	30	1.9167	1	1	1
69	LVAI	60	1.8153	1	1	1
70	WAVV	60	2.13	1	1	1
71	MMLL	90	1.8547	1	1	1
72	VALV	120	1.8622	1	1	1
73	CFLW	120	2.3126	1	1	1
74	WWCP	120	2.2588	1	1	1
75	IWPW	120	2.3147	1	1	1
76	WPYY	120	2.3539	1	1	1
77	VSWY	120	2.2386	1	1	1
78	SWYF	120	2.2834	1	1	1
79	WWMF	120	2.324	1	1	1
80	SYFF	30	2.2976	1	1	1
81	WFMM	30	2.277	1	1	1
82	CWFS	30	2.2304	1	1	1
83	PWYW	120	2.3582	1	1	1
84	LRFH	120	2.0365	1	1	1
85	KVWL	60	2.0713	1	1	1
86	FYAY	120	2.2045	1	1	1
87	MMYY	120	2.2698	1	1	1
88	KLLF	30	2.0251	1	1	1

89	SAWF	60	2.205	1	1	1
90	TGFF	120	2.186	1	1	1
91	VAFY	120	2.261	1	1	1
92	YAYY	120	2.2018	1	1	1
93	PYYI	120	2.2951	1	1	1
94	YPYI	120	2.257	1	1	1
95	PGWW	120	2.2539	1	1	1
96	VYTF	120	2.2294	1	1	1
97	SSFF	30	2.1993	1	1	1
98	CTYF	30	2.2442	1	1	1
99	FSYV	60	2.2074	1	1	1
100	GVLF	60	2.0922	1	1	1
101	STYF	60	2.1849	1	1	1
102	GLLY	120	2.0518	1	1	1
103	ILVV	30	2.0299	1	1	1
104	IVII	30	2.0101	1	1	1
105	LVII	30	1.9902	1	1	1
106	LVLII	30	1.9717	1	1	1
107	SLVV	60	1.9995	1	1	1
108	VILI	60	2.0007	1	1	1
109	VLVL	120	2.0006	1	1	1
110	GVFY	30	2.1954	1	1	1
111	IMVV	30	1.9851	1	1	1
112	LTLF	30	2.1648	1	1	1
113	LTLI	30	1.9326	1	1	1
114	LWFF	30	2.3831	1	1	1
115	IQII	60	1.8	1	1	1
116	LVIF	60	2.2381	1	1	1
117	LVWF	90	2.3499	1	1	1
118	WVII	60	2.2171	1	1	1
119	LVIY	120	2.2135	1	1	1
120	KMLF	120	2.005	1	1	1
121	LHII	120	2.0624	1	1	1
122	QHII	120	1.8795	1	1	1
123	QMVV	120	1.7315	1	1	1
124	WTIF	30	2.2476	1	1	1
125	IVIY	60	2.2224	1	1	1
126	NTIF	60	2.0149	1	1	1
127	IWII	120	2.2275	1	1	1
128	LTWF	120	2.2806	1	1	1
129	LWFI	120	2.3355	1	1	1
130	WVFI	120	2.3253	1	1	1
131	FVIY	30	2.2852	1	1	1
132	LLLF	30	2.1976	1	1	1
133	LVHF	30	2.217	1	1	1
134	PIFY	30	2.371	1	1	1
135	WIIF	30	2.3252	1	1	1
136	IFIY	60	2.3126	1	1	1
137	IVFW	120	2.3632	1	1	1
138	VLVV	120	2.0266	1	1	1

139	HLLF	90	2.1426	1	1	1
140	IHFV	120	2.2026	1	1	1
141	IIIR	120	1.7045	1	1	1
142	LVPF	120	2.2222	1	1	1
143	IVIF	30	2.2498	1	1	1
144	LLHF	30	2.2154	1	1	1
145	LLLY	30	2.1815	1	1	1
146	CIYF	60	2.331	1	1	1
147	PVFW	60	2.3756	1	1	1
148	PVII	90	1.9847	1	1	1
149	PWYY	60	2.3551	1	1	1
150	WAFY	60	2.29	1	1	1
151	FHLF	30	2.2078	1	1	1
152	AVIQ	120	1.6139	1	1	1
153	GVNM	120	1.4004	1	1	1
154	LALR	120	1.4103	1	1	1
155	LIEM	90	1.4768	1	1	1
156	LLRQ	120	1.3627	1	0	1
157	PMFY	60	2.3503	1	1	1
158	CAYW	120	2.2327	1	1	1
159	FAFP	120	2.22	1	1	1
160	FMYS	120	2.2133	1	1	1
161	KFLL	120	1.9793	1	1	1
162	PWWC	120	2.3162	1	1	1
163	VFGV	120	2.0687	1	1	1
164	FCWW	120	2.331	1	1	1
165	TRFS	120	1.9402	1	1	1

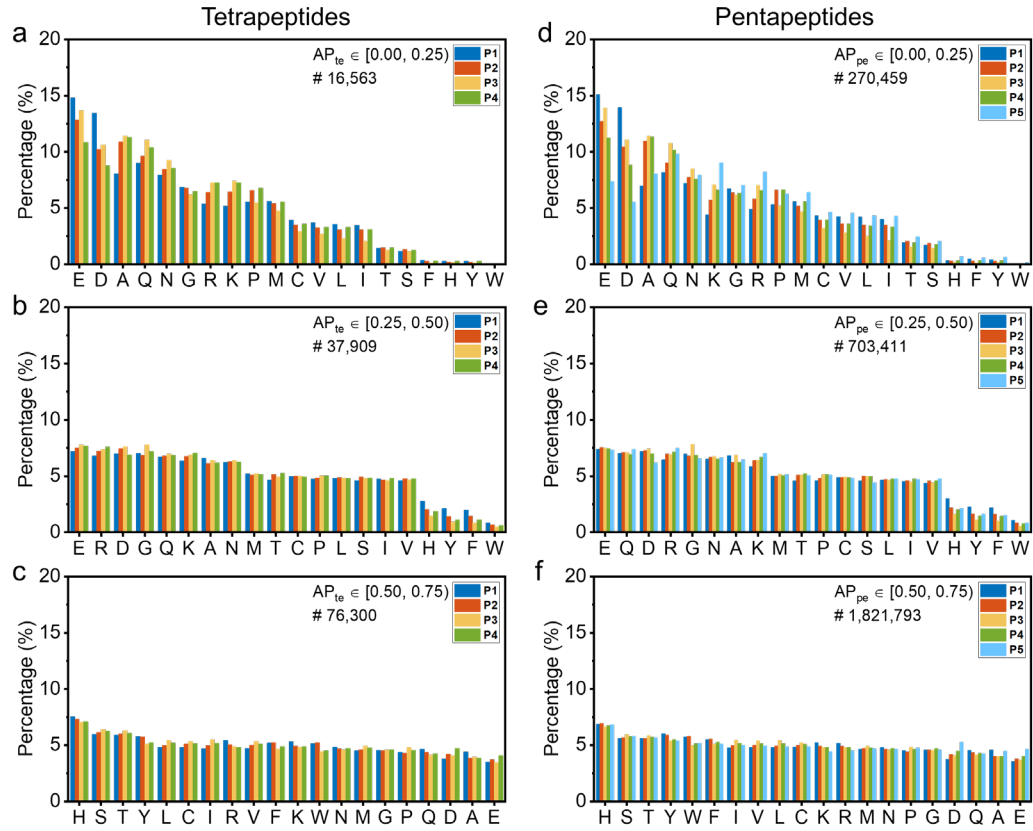


FIGURE S2 a-c Percentage of each amino acid at each position within the range of [0.00, 0.25), [0.25, 0.50), [0.50, 0.75) of tetrapeptides. **d-f** Percentage of each amino acid at each position within the range of [0.00, 0.25), [0.25, 0.50), [0.50, 0.75) of pentapeptides.

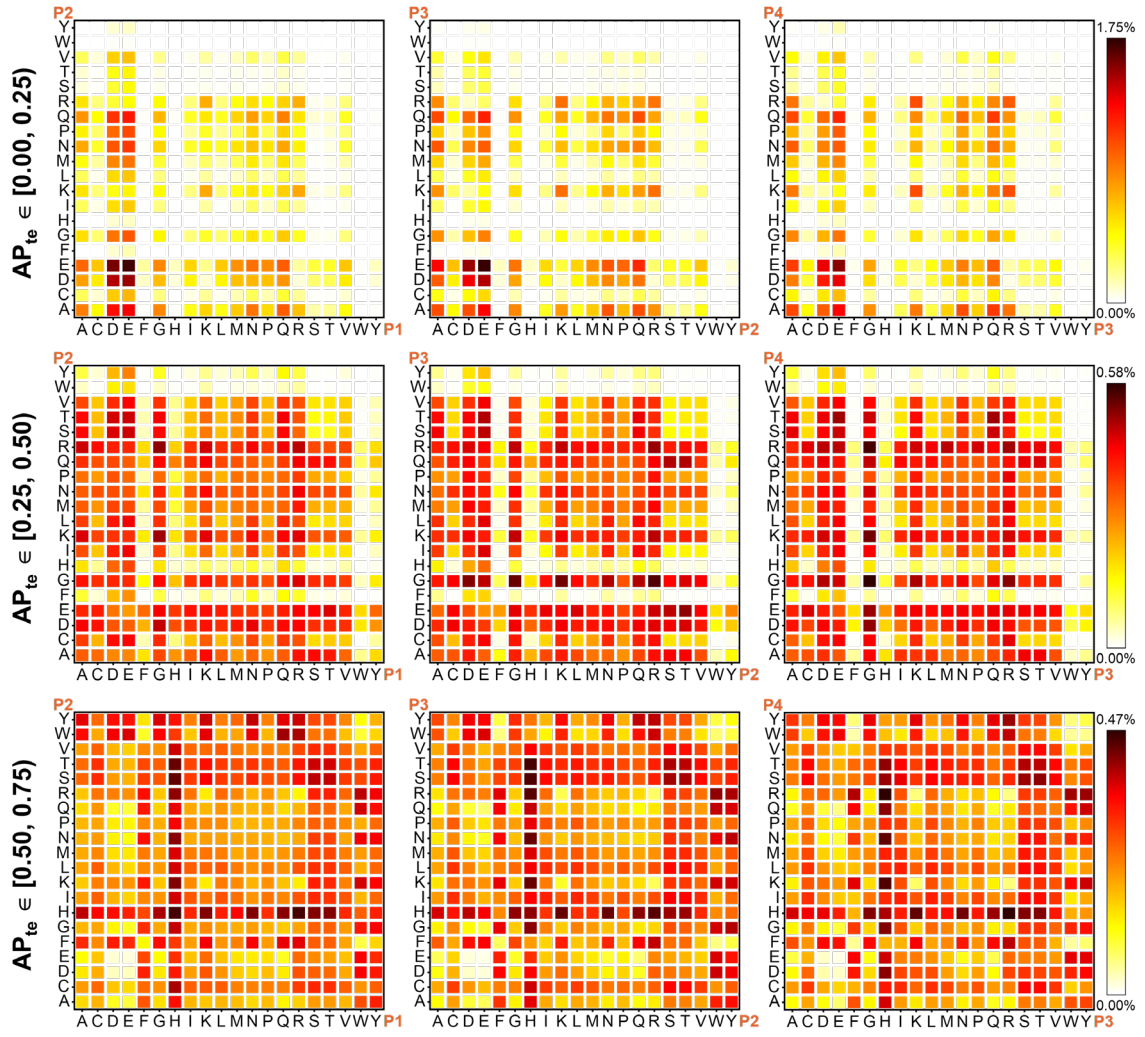


FIGURE S3 Percentage of amino acid pairs at each position within the range of $A \in [0.00, 0.25)$, $B \in [0.25, 0.50)$, $C \in [0.50, 0.75)$ of tetrapeptides.

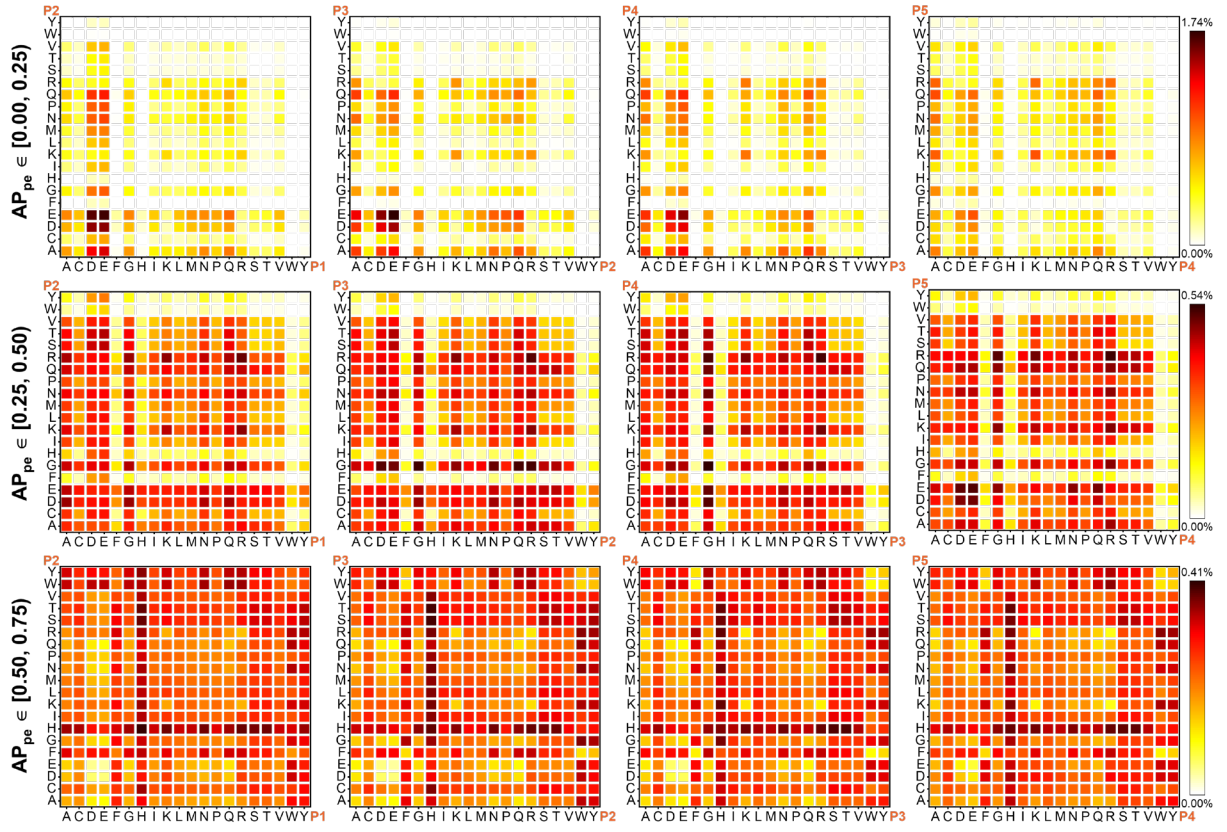


FIGURE S4 Percentage of amino acid pairs at each position within the range of $A \in [0.00, 0.25)$, $B \in [0.25, 0.50)$, $C \in [0.50, 0.75)$ of pentapeptides.

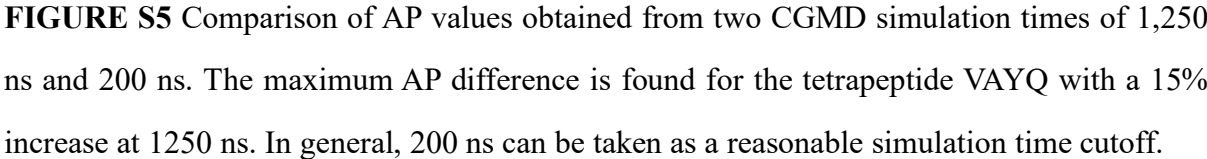


Table S2: AP ranking of different groups. A smaller ranking indicates a smaller AP. The “-” symbol between the peptides indicates “minus” sign.

Peptide	Rank difference	Peptide	Rank difference	Peptide	Rank difference
NNFN-NNHN	6745	NNNF-NNNH	10057	NNFF-NNHH	-34576
NFNN-NHNN	10780	NNFN-NNHN	6745	NFFN-NHHN	-32761
FNNN-HNNN	14454	NFNN-NHNN	10780	FFNN-HHNN	-30303
NNYN-NNHN	4119	NNNY-NNNH	6405	NNYY-NNHH	-28579
NYNN-NHNN	4706	NNYN-NNHN	4119	NYYN-NHHN	-26462
HHNH-HHKK	3868	HHHN-HHKK	248	HHNN-HHKK	-26227
KKFK-KHKK	1492	KKKF-KKKH	1872	KKFF-KKHH	-25825
FKKK-HKKK	2447	KFKK-KHKK	3267	FFKK-HHKK	-25445
YKKK-HKKK	904	KYKK-KHKK	1602	YYKK-HHKK	-23332
KFKK-KHKK	3267	KKFK-KHKK	1492	KFFK-KHHK	-22579
YNNN-HNNN	7128	NYNN-NHNN	4706	YYNN-HHNN	-21273
TGTT-TKTT	1337	TTGT-TTKT	2387	TGGT-TKKT	-20095
HNNH-HKHH	2972	HHNH-HHKK	3868	HNNH-HKKH	-19298
FDDD-HDDD	1661	DFDD-DHDD	683	FFDD-HHDD	-19001
EEFE-EEHE	1306	EEEF-EEEH	1304	EEFF-EEHH	-16972
FQQQ-YQQQ	1774	QFQQ-QYQQ	1098	FFQQ-YYQQ	-15531
NYYY-KYYY	669	YNY-YKYY	13	NNYY-KKYY	-15476
EFEE-EHEE	1464	EEFE-EEHE	1306	EEFE-EHHE	-15247
RRFR-RRHR	5068	RRRF-RRRH	4927	RRFF-RRHH	-15024
RRYR-RRHR	2672	RRRY-RRRH	3146	RRYY-RRHH	-14989
RFRR-RHRR	5056	RRFR-RRHR	5068	RFFR-RHHR	-14911
NIII-KIII	3973	INII-IKII	1323	NNII-KKII	-14555
TQTT-TKTT	2547	TTQT-TTKT	1875	TQQT-TKKT	-14007
QIII-KIII	5603	IQII-IKII	589	QQII-KKII	-13867
EYEE-EHEE	862	EEYE-EEHE	1148	EYYE-EHHE	-13854
RYRR-RHRR	2872	RRYR-RRHR	2672	RYR-RHHR	-13461
NHHH-KHHH	6186	HHNH-HKHH	2972	NNHH-KKHH	-13342
EEYE-EEHE	1148	EEY-EEEH	1055	EEYY-EEHH	-12788
GTTT-KTTT	5738	TGTT-TKTT	1337	GGTT-KKTT	-12469
FRRR-HRRR	4242	RFRR-RHRR	5056	FFRR-HHRR	-11773
TTGT-TTAT	1088	TTTG-TTTA	2271	TTGG-TTAA	-11582
YRRR-HRRR	2352	RYRR-RHRR	2872	YYRR-HHRR	-11527
INNN-GNNN	2998	NINN-NGNN	884	IINN-GGNN	-11481
HHGH-HHRH	9589	HHHG-HHHR	4365	HHGG-HHRR	-11282
WRRW-WDDW	1402	WRRW-WDDW	1424	WRRW-WDDW	-11138
TGTT-TATT	1885	TTGT-TTAT	1088	TGGT-TAAT	-10171
QTTT-KTTT	4618	TQTT-TKTT	2547	QQTT-KKTT	-9709
EEEE-HEEE	496	EREE-EHEE	2172	RREE-HHEE	-9119
FNNN-YNNN	7326	NFNN-NYNN	6074	FFNN-YYNN	-9030

NPNN-NDNN	2837	NNPN-NNDN	2403	NPPN-NDDN	-8980
CCCC-CCSC	460	CCCC-CCCS	1099	CCCC-CCSS	-8800
QQPQ-QQRQ	257	QQQP-QQQR	1345	QQPP-QQRR	-8690
VIII-HIII	1259	IVII-IHII	10187	VVII-HHII	-8560
HHGH-HHKH	13235	HHHG-HHHK	8251	HHGG-HHKK	-8428
GIGG-GTGG	3086	GGIG-GGTG	153	GIIG-GTTG	-8384
LSSS-HSSS	180	SLSS-SHSS	5252	LLSS-HHSS	-8100
QPQQ-QRQQ	2187	QQPQ-QQRQ	257	QPPQ-QRRQ	-7976
VVVV-HVVV	1089	VVVV-VHVV	8851	VVVV-HHVV	-7937
YYQY-YYKY	3741	YYYQ-YYYK	1533	YYQQ-YYKK	-7914
YEEE-HEEE	1173	EYEE-EHEE	862	YYEE-HHEE	-7772
QCCC-KCCC	6921	CQCC-CKCC	1733	QQCC-KKCC	-7727
IQII-IKII	589	IIQI-IIKI	1082	IQQI-IKKI	-7706
CCCC-CSCC	1325	CCCC-CCSC	460	CCCC-CSSC	-7597
HHAH-HHRH	8831	HHHA-HHHR	3407	HHAA-HHRR	-7592
EEEE-HEEE	1833	EFEE-EHEE	1464	FFEE-HHEE	-7471
GGVG-GGTG	2247	GGGV-GGGT	3011	GGVV-GGTT	-7416
ICII-ITII	42	IICI-IITI	43	ICCI-ITTI	-7083
DDFD-DDYD	239	DDDF-DDDY	666	DDFF-DDYY	-6888
VCCC-HCCC	5796	CVCC-CHCC	14779	VVCC-HHCC	-6871
NPNN-NKNN	5519	NNPN-NNKN	2116	NPPN-NKKN	-6649
NNPN-NNDN	2403	NNNP-NNND	7701	NNPP-NNDD	-6634
YQYY-YKYY	2855	YYQY-YYKY	3741	YQQY-YKKY	-6506
DFDD-DYDD	1009	DDFD-DDYD	239	DDFF-DYYD	-6483
NNPN-NNKN	2116	NNNP-NNNK	3970	NNPP-NNKK	-6462
HAHH-HRHH	4183	HHAH-HHRH	8831	HAAH-HRRH	-6303
GGIG-GGTG	153	GGGI-GGGT	2567	GGII-GGTT	-6301
NFNN-NYNN	6074	NNFN-NNYN	2626	NFFN-NYYN	-6299
NPNN-NRNN	5686	NNPN-NNRN	1638	NPPN-NRRN	-6178
NNPN-NNRN	1638	NNNP-NNNR	3642	NNPP-NNRR	-6107
GIGG-GSGG	3924	GGIG-GGSG	748	GIIG-GSSG	-6069
IICI-IISI	850	IIIC-IIIS	763	IICC-IISS	-6002
NNFN-NNYN	2626	NNNF-NNNY	3652	NNFF-NNYY	-5997
FDDD-RDDD	7413	DFDD-DRDD	960	FFDD-RRDD	-5977
NNPN-NQNN	715	NNNP-NNNQ	2584	NNPP-NNQQ	-5946
NPNN-NQNN	4376	NNPN-NQNN	715	NPPN-NQQN	-5783
GGVG-GGSG	2842	GGGV-GGGS	4214	GGVV-GGSS	-5710
NMNN-NNNN	3593	NNMN-NNNN	593	NMMN-NNNN	-5555
HGHH-HRHH	6047	HHGH-HHRH	9589	HGGH-HRRH	-5527
AAGA-AAKA	687	AAAG-AAAK	821	AAGG-AAKK	-5433
GVGG-GTGG	3599	GGVG-GGTG	2247	GVVG-GTTG	-5312
ICII-ISII	748	IICI-IISI	850	ICCI-ISSI	-5050
NNMN-NNNN	593	NNNM-NNNN	3903	NNMM-NNNN	-5041

GGLG-GGTG	1313	GGGL-GGGT	2871	GGLL-GGTT	-4927
AHHH-RHHH	9838	HAHH-HRHH	4183	AAHH-RRHH	-4912
AGAA-AKAA	1480	AAGA-AAKA	687	AGGA-AKKA	-4805
GSSS-ASSS	365	SGSS-SASS	260	GGSS-AASS	-4789
GLGG-GTGG	3941	GGLG-GGTG	1313	GLLG-GTTG	-4774
HHAH-HHKH	12477	HHHA-HHHK	7293	HHAA-HHKK	-4738
FDDD-YDDD	1236	DFDD-DYDD	1009	FFDD-YYDD	-4633
GGIG-GGSG	748	GGGI-GGGS	3770	GGII-GGSS	-4595
QYYY-KYYY	3907	YQYY-YKYY	2855	QQYY-KKYY	-4591
QQNQ-QQKQ	627	QQQN-QQQK	253	QQNN-QQKK	-4576
NINN-NCNN	5285	NNIN-NNCN	2967	NIIN-NCCN	-4538
NINN-NVNN	3252	NNIN-NNVN	754	NIIN-NVVN	-4473
IAAA-TAAA	2628	AIAA-ATAA	2326	IIAA-TTAA	-4458
EEEE-HEEE	786	EKEE-EHEE	2823	KKEE-HHEE	-4429
SRSS-SKSS	1239	SSRS-SSKS	878	SRRS-SKKS	-4291
PPCP-PPTP	1977	PPPC-PPPT	751	PPCC-PPTT	-4262
EEFE-EEYE	158	EEEF-EEY	249	EEFF-EEYY	-4184
IAAA-SAAA	3664	AIAA-ASAA	1261	IIAA-SSAA	-4182
QMQQ-QGQQ	342	QMQQ-QQGQ	343	QMMQ-QGGQ	-4170
MQQQ-KQQQ	2446	QMQQ-QKQQ	814	MMQQ-KKQQ	-4156
PPLP-PPTP	156	PPPL-PPPT	125	PPLL-PPTT	-4102
TQTT-TATT	3095	TTQT-TTAT	576	TQQT-TAAT	-4083
LLCL-LLSL	3091	LLLC-LLLS	2505	LLCC-LLSS	-4083
NNIN-NNVN	754	NNNI-NNNV	3169	NNII-NNVV	-3878
QQNQ-QQRQ	1012	QQQN-QQQR	492	QQNN-QQRR	-3875
LTTT-HTTT	2045	TLTT-THTT	7326	LLTT-HHTT	-3848
SSVS-SSLS	201	SSSV-SSSL	722	SSVV-SSLL	-3758
HHGH-HHAH	758	HHHG-HHHA	958	HHGG-HHAA	-3690
SSRS-SSKS	878	SSSR-SSSK	948	SSRR-SSKK	-3676
IILI-IISI	569	IIIL-IIIS	2329	IILL-IISS	-3646
HWWW-GWWW	918	WHWW-WGWW	840	HHWW-GGWW	-3629
NLNN-NCNN	4055	NNLN-NNCN	1817	NLLN-NCCN	-3625
PQQQ-KQQQ	2334	QPQQ-QKQQ	1782	PPQQ-KKQQ	-3621
SISS-SHSS	1918	SSIS-SSHS	8813	SIIS-SHHS	-3609
AAVA-AASA	5452	AAAV-AAAS	6491	AAVV-AASS	-3580
AAVA-AATA	5949	AAAV-AAAT	5913	AAVV-AATT	-3575
IGGG-TGGG	7941	GIGG-GTGG	3086	IIGG-TTGG	-3556
AFAA-AWAA	14467	AAFA-AAWA	13992	AFFA-AWWA	-3478
GHHH-RHHH	11329	HGHH-HRHH	6047	GGHH-RRHH	-3462
HHQH-HHRH	10105	HHHQ-HHHR	6607	HHQQ-HHRR	-3418
MQQQ-RQQQ	2371	QMQQ-QRQQ	1219	MMQQ-RRQQ	-3404
WWRW-WWDW	1424	WWWR-WWWD	3724	WWRR-WWDD	-3393
NNIN-NNCN	2967	NNNI-NNNC	4946	NNII-NNCC	-3289

CCCC-SCCC	3534	CCCC-CSCC	1325	CCCC-SSCC	-3231
GGLG-GGSG	1908	GGGL-GGGS	4074	GGLL-GGSS	-3221
FTTT-WTTT	4512	TFTT-TWTT	900	FFTT-WWTT	-3200
GTTT-ATTT	2517	TGTT-TATT	1885	GGTT-AATT	-3078
LLLL-LTLL	957	LLLL-LLTL	3764	LLLL-LTTL	-3064
SVSS-SLSS	121	SSVS-SSLS	201	SVVS-SLLS	-3045
GVVG-GSGG	4437	GGVG-GGSG	2842	GVVG-GSSG	-2997
RTRR-RERR	381	RRTR-RRER	370	RTTR-REER	-2947
CCQC-CCKC	5610	CCCQ-CCCK	1020	CCQQ-CCKK	-2912
PQQQ-RQQQ	2259	QPQQ-QRQQ	2187	PPQQ-RRQQ	-2869
CIII-TIII	3823	ICII-ITII	42	CCII-TTII	-2866
HFFF-GFFF	889	FHFF-FGFF	932	HHFF-GGFF	-2828
TTQT-TTAT	576	TTTQ-TTTA	3163	TTQQ-TTAA	-2791
YYHY-YYTY	127	YYYH-YYYT	1332	YYHH-YYTT	-2770
PCPP-PTPP	1231	PPCP-PPTP	1977	PCCP-PTTP	-2690
QQMQ-QQGQ	343	QQQM-QQQG	1148	QQMM-QQGG	-2647
QNQQ-QKQQ	1508	QQNQ-QQKQ	627	QNNQ-QKKQ	-2593
NNLN-NNCN	1817	NNNL-NNNC	3723	NNLL-NNCC	-2550
GGVG-GGLG	934	GGGV-GGGL	140	GGVV-GGLL	-2489
VAAA-TAAA	4335	AVAA-ATAA	5575	VVAA-TTAA	-2483
CQCC-CKCC	1733	CCQC-CCKC	5610	CQQC-CKKC	-2480
LCLL-LSLL	3644	LLCL-LLSL	3091	LCCL-LSSL	-2470
LLLL-LLTL	3764	LLLL-LLLT	4268	LLLL-LLTT	-2462
GLGG-GSGG	4779	GGLG-GGSG	1908	GLLG-GSSG	-2459
CCNC-CCAC	421	CCCN-CCCA	4029	CCNN-CCAA	-2427
PSSS-HSSS	1608	SPSS-SHSS	7365	PPSS-HHSS	-2426
HYYY-AYYY	1229	YHYY-YAYY	2079	HHYY-AAYY	-2322
HFHH-HWHH	441	HHFH-HHWH	342	HFFH-HWWH	-2321
GFGG-GWGG	11158	GGFG-GWGW	8817	GFFG-GWWG	-2290
FSSS-WSSS	3583	SFSS-SWSS	1173	FFSS-WWSS	-2219
VAAA-SAAA	5371	AVAA-ASAA	4510	VVAA-SSAA	-2207
LQQQ-VQQQ	519	QLQQ-QVQQ	34	LLQQ-VVQQ	-2196
AVAA-ASAA	4510	AAVA-AASA	5452	AVVA-ASSA	-2167
QPQQ-QGQQ	1310	QQPQ-QQGQ	1450	QPPQ-QGGQ	-2150
FKKK-YKKK	1543	KFKK-KYKK	1665	FFKK-YYKK	-2113
GQQQ-KQQQ	2373	QGQQ-QKQQ	472	GGQQ-KKQQ	-2091
MQQQ-GQQQ	73	QMQQ-QGQQ	342	MMQQ-GGQQ	-2065
LGGG-VGGG	956	GLGG-GVGG	342	LLGG-VVGG	-2065
LGGG-TGGG	7556	GLGG-GTGG	3941	LLGG-TTGG	-1997
HAHH-HKHH	7694	HHAH-HHKH	12477	HAAH-HKKH	-1960
AALA-AASA	2770	AAAL-AAAS	6124	AALL-AASS	-1950
AALA-AATA	3267	AAAL-AAAT	5546	AALL-AATT	-1945
GGFG-GGWG	8817	GGGF-GGGW	14988	GGFF-GGWW	-1910

LVVV-HVVV	5053	VLVV-VHVV	10393	LLVV-HHVV	-1901
MMCM-MMSM	1946	MMMC-MMMS	2860	MMCC-MMSS	-1654
QNQQ-QRQQ	1913	QQNQ-QQRQ	1012	QNNQ-QRRQ	-1651
AAVA-AALA	2682	AAAV-AAAL	367	AAVV-AALL	-1630
ANAA-ADAA	23	AANA-AADA	192	ANNA-ADDA	-1577
ICCC-HCCC	9892	CICC-CHCC	17319	IICC-HHCC	-1570
AANA-AADA	192	AAAN-AAAD	600	AANN-AADD	-1544
AVAA-ALAA	2349	AAVA-AALA	2682	AVVA-ALLA	-1543
TRTT-TKTT	2658	TTRT-TTKT	2313	TRRT-TKKT	-1523
RSSS-KSSS	1488	SRSS-SKSS	1239	RRSS-KKSS	-1521
LAAA-TAAA	3061	ALAA-ATAA	3226	LLAA-TTAA	-1518
AANA-AAAA	304	AAAN-AAAA	158	AANN-AAAA	-1480
WWHW-WWTW	234	WWWH-WWWT	1001	WWHH-WWTT	-1477
RFRR-RYRR	2184	RRFR-RRYR	2396	RFFR-RYYR	-1450
I IPI-IIMI	433	I IIP-IIIM	310	I IPP-I IMM	-1412
EFEE-EYEE	602	EEFE-EEYE	158	EFFE-EYYE	-1393
MCM-MMSM	1221	MMCM-MMSM	1946	MCCM-MSSM	-1385
PNNN-KNNN	6762	NPNN-NKNN	5519	PPNN-KKNN	-1377
AAFA-AAWA	13992	AAAF-AAAW	20735	AAFF-AAWW	-1364
CNCC-CACC	4320	CCNC-CCAC	421	CNNC-CAAC	-1344
GQQQ-RQQQ	2298	QGQQ-QRQQ	877	GGQQ-RRQQ	-1339
TTVT-TTIT	1456	TTTV-TTTI	197	TTVV-TTII	-1320
PPNP-PPKP	7681	PPPN-PPPK	424	PPNN-PPKK	-1314
TTRT-TTKT	2313	TTTR-TTTK	1744	TTRR-TTKK	-1257
YYCY-YYMY	180	YYYC-YYYM	21	YYCC-YYMM	-1249
LAAA-SAAA	4097	ALAA-ASAA	2161	LLAA-SSAA	-1242
PNNN-QNNN	2097	NPNN-NQNN	4376	PPNN-QQNN	-1235
KPKK-KGKK	544	KKPK-KKGK	189	KPPK-KGGK	-1210
QGGG-KGGG	5076	GQGG-GKGG	2533	QQGG-KKGG	-1198
HGHH-HKHH	9558	HHGH-HHKH	13235	HGGH-HKKH	-1184
ANAA-AAAA	306	AANA-AAAA	304	ANNA-AAAA	-1152
WWWW-WWLW	12	WWWW-WWWL	257	WWWW-WWLL	-1135
GGVG-GGIG	2094	GGGV-GGGI	444	GGVV-GGII	-1115
HHFH-HHWH	342	HHHF-HHHW	2368	HHFF-HHWW	-1109
MRMM-MQMM	3883	MMRM-MMQM	1301	MRRM-MQQM	-1093
PPNP-PPQP	7268	PPPN-PPPQ	3634	PPNN-PPQQ	-1093
WWWW-WWPW	304	WWWW-WWWP	280	WWWW-WWPP	-1077
AVAA-ATAA	5575	AAVA-AATA	5949	AVVA-ATTA	-1051
PDDD-QDDD	15	DPDD-DQDD	44	PPDD-QQDD	-1030
NCCC-ACCC	10140	CNCC-CACC	4320	NNCC-AACC	-1026
VAAA-LAAA	1274	AVAA-ALAA	2349	VVAA-LLAA	-965
LLLL-TLLL	4298	LLLL-LTLL	957	LLLL-TTLL	-964
WWWW-WWIW	53	WWWW-WWWI	82	WWWW-WWII	-957

PNNN-RNNN	5868	NPNN-NRNN	5686	PPNN-RRNN	-916
NINN-NLNN	1230	NNIN-NNLN	1150	NIIN-NLLN	-913
TITT-THTT	4350	TTIT-TTHT	10374	TIIT-THHT	-903
HQHH-HRHH	8216	HHQH-HHRH	10105	HQQH-HRRH	-900
INNN-VNNN	4291	NINN-NVNN	3252	IINN-VVNN	-878
NQNN-NKNN	1143	NNQN-NNKN	1401	NQQN-NKKN	-866
CFCC-CWCC	2076	CCFC-CCWC	1708	CFFC-CWWC	-863
IGGG-SGGG	8205	GIGG-GSGG	3924	IIGG-SSGG	-832
VFVV-VWVV	971	VVFF-VVWV	464	VFFF-VVWV	-825
RVRR-RCRR	29	RRVR-RRCR	166	RVVR-RCCR	-824
CMCC-CTCC	2702	CCMC-CCTC	1801	CMMC-CTTC	-811
VRVV-VKVV	8949	VVRV-VVKV	6918	VRRV-VKKV	-802
YYIY-YYPY	193	YYII-YYYP	104	YYII-YYPP	-798
FHHH-WHHH	2826	HFHH-HWHH	441	FFHH-WWHH	-788
IDDD-LDDD	9	DIDD-DLDD	3	IIDD-LLDD	-765
QGGG-RGGG	4491	GQGG-GRGG	2545	QQGG-RRGG	-763
NNIN-NNLN	1150	NNNI-NNNL	1223	NNII-NNLL	-739
DVDD-DCDD	43	DDVD-DDCD	50	DVVD-DCCD	-726
CCFC-CCWC	1708	CCCF-CCCW	2592	CCFF-CCWW	-709
LNNN-VNNN	2908	NLNN-NVNN	2022	LLNN-VVNN	-705
IFII-IWII	1992	IIFI-IIWI	827	IFFI-IWWI	-694
NNQN-NNDN	1688	NNNQ-NNND	5117	NNQQ-NNDD	-688
VDDD-CDDD	148	DVDD-DCDD	43	VVDD-CCDD	-674
QQQQ-QQDQ	569	QQQQ-QQQD	2279	QQQQ-QQDD	-670
VVFF-VVWV	464	VVVF-VVWV	783	VVFF-VVWW	-653
ALAA-ASAA	2161	AALA-AASA	2770	ALLA-ASSA	-624
VSVV-VTVV	578	VVSV-VVTV	698	VSSV-VTTV	-613
CRCC-CKCC	5964	CCRC-CCKC	6383	CRRC-CKKC	-612
FFCF-FFMF	81	FFFC-FFFM	85	FFCC-FFMM	-594
NQQQ-KQQQ	2638	QNQQ-QKQQ	1508	NNQQ-KKQQ	-590
VLLL-HLLL	10311	LVLL-LHLL	20044	VVLL-HHLL	-586
FCCC-WCCC	4346	CFCC-CWCC	2076	FFCC-WWCC	-572
LFLL-LWLL	3684	LLFL-LLWL	2305	LFLL-LWWL	-565
HHQH-HHKH	13751	HHHQ-HHKK	10493	HHQQ-HHKK	-564
NNRN-NNDN	765	NNNR-NNND	4059	NNRR-NNDD	-527
EVVE-ECEE	18	EEVE-EECE	42	EVVE-ECCE	-521
RRTR-RRER	370	RRRT-RRRE	2862	RRTT-RREE	-519
NNQN-NNKN	1401	NNNQ-NNNK	1386	NNQQ-NNKK	-516
RRIR-RRLR	64	RRRI-RRRL	194	RRII-RRLL	-512
GAAA-RAAA	1327	AGAA-ARAA	700	GGAA-RAAA	-497
PNPP-PQPP	5598	PPNP-PPQP	7268	PNNP-PQQP	-490
MDDD-GDDD	67	DMDD-DGDD	36	MMDD-GGDD	-470
VEEE-CEEE	52	EVVE-ECEE	18	VVEE-CEEE	-457

PFPP-PWPP	3968	PPFP-PPWP	2288	PFPP-PWWP	-454
MMRM-MMQM	1301	MMMR-MMMQ	6738	MMRR-MMQQ	-444
FVVV-WVVV	2885	VFVV-VWVV	971	FFVV-WWVV	-441
DDVD-DDCD	50	DDDV-DDDC	18	DDVV-DDCC	-415
NQNN-NRNN	1310	NNQN-NNRN	923	NQQN-NRRN	-395
SVVV-TVVV	1163	VSVV-VTVV	578	SSVV-TTVV	-395
KFKK-KYKK	1665	KKFK-KKYK	1545	KFFK-KYYK	-393
IKKK-LKKK	201	KIKK-KLKK	537	I IKK-LLKK	-383
LLFL-LLWL	2305	LLLF-LLLW	2776	LLFF-LLWW	-373
FIII-WIII	3706	IFII-IWII	1992	FFII-WWII	-372
KKPK-KKGK	189	KKKP-KKKG	877	KKPP-KKGG	-370
FFVF-FFPF	247	FFFV-FFFP	70	FFVV-FFPP	-369
PRRR-GRRR	15	RPRR-RGRR	741	PPRR-GGRR	-361
VVVV-VVIV	2306	VVVV-VVVI	1489	VVVV-VVII	-360
NNRN-NNKN	478	NNNR-NNNK	328	NNRR-NNKK	-355
CPPP-TPPP	2844	PCPP-PTPP	1231	CCPP-TTPP	-341
CCRC-CCKC	6383	CCCR-CCCK	6018	CCRR-CCKK	-321
QTTT-ATTT	1397	TQTT-TATT	3095	QQT-TAATT	-318
DMDD-DGDD	36	DDMD-DDGD	100	DMMD-DGGD	-302
FFFF-LFFF	32	FFFF-FLFF	2	FFFF-LLFF	-299
IIFI-IIWI	827	IIIF-IIIW	1167	I IFF-IIWW	-299
WWW-CWWW	140	WWW-WCWW	158	WWW-CCWW	-284
FFVF-FFLF	183	FFFV-FFFL	104	FFVV-FFLL	-279
GGRG-GGKG	291	GGGR-GGGK	409	GGRR-GGKK	-274
PNPP-PKPP	5484	PPNP-PPKP	7681	PNNP-PKKP	-266
GAGG-GDGG	1567	GGAG-GGDG	1740	GAAG-GDDG	-264
VVRV-VVKV	6918	VVVR-VVVK	8075	VVRR-VVKK	-263
FRRR-YRRR	1890	RFRR-RYRR	2184	FFRR-YYRR	-246
NAAA-QAAA	123	ANAA-AQAA	109	NNAA-QQAA	-243
YYVY-YYLY	473	YYYV-YYYL	407	YYVV-YYLL	-229
EEEE-QEEE	31	EPEE-EQEE	39	PPEE-QQEE	-225
KKTK-KKSK	418	KKKT-KKKS	37	KKTT-KKSS	-220
WWVW-WWMW	223	WWWV-WWWM	121	WWVV-WWMM	-199
INNN-LNNN	1383	NINN-NLNN	1230	I INN-LLNN	-173
NNKN-NNDN	287	NNNK-NNND	3731	NNKK-NNDD	-172
WWW-WLWW	876	WWW-WLW	12	WWW-WLLW	-171
KKAK-KKKK	59	KKKA-KKKK	152	KKAA-KKKK	-170
NNQN-NNRN	923	NNNQ-NNNR	1058	NNQQ-NNRR	-161
PPCP-PPLP	1821	PPPC-PPPL	626	PPCC-PPLL	-160
MFMM-MWMM	7396	MMFM-MMWM	6150	MFFM-MWWM	-145
QNNN-KNNN	4665	NQNN-NKNN	1143	QQNN-KKNN	-142
IRRR-CRRR	353	RIRR-RCRR	420	I IRR-CCRR	-138
YYYY-MYYY	637	YYYY-YMY	50	YYYY-MMYY	-138

YYVY-YPY	839	YYYV-YYYY	370	YYVV-YYPP	-136
AQAA-AAAA	197	AAQA-AAAA	377	AQQA-AAAA	-129
WWIW-WWPW	251	WWWI-WWWP	198	WWII-WWPP	-120
EEME-EEGE	17	EEEM-EEEG	32	EEMM-EEGG	-112
QHHH-RHHH	11764	HQHH-HRHH	8216	QQHH-RRHH	-109
CCMC-CCTC	1801	CCCM-CCCT	2044	CCMM-CCTT	-101
PPFP-PPWP	2288	PPPF-PPPW	2800	PPFF-PPWW	-100
FIFF-FPFF	36	FFIF-FFPF	15	FIIF-FPPF	-97
HHCH-HHMH	1684	HHHC-HHMH	741	HHCC-HHMM	-97
QQPQ-QQGQ	1450	QQQP-QQQG	1999	QQPP-QQGG	-92
FFFF-FFWF	3	FFFF-FFFW	28	FFFF-FFWW	-88
DPDD-DNDD	219	DDPD-DDND	79	DPPD-DNND	-84
KIKK-KEKK	1909	KKIK-KKEK	1055	KIIK-KEEK	-84
FLLL-WLLL	4516	LFLl-LWLL	3684	FLLL-WWLL	-75
NVNN-NCNN	2033	NNVN-NNCN	2213	NVVN-NCCN	-65
FFFF-IFFF	56	FFFF-FIFF	22	FFFF-IIFF	-45
GAAA-KAAA	2308	AGAA-AKAA	1480	GGAA-KKAA	-44
PRPP-PQPP	7543	PPRP-PPQP	5068	PRRP-PQQP	-37
AAQA-AADA	265	AAAQ-AAAD	761	AAQQ-AADD	-37
EEVE-EECE	42	EEEV-EEEC	25	EEVV-EECC	-36
RRFR-RRYR	2396	RRRF-RRRY	1781	RRFF-RRYY	-35
QAAQ-QEQQ	403	QQAQ-QQEQ	583	QAAQ-QEEQ	-31
KKAK-KKRK	109	KKKA-KKKR	166	KKAA-KKRR	-12
MRMM-MKMM	6607	MMRM-MMKM	5181	MRRM-MKKM	-10
VFFF-LFFF	18	FVFF-FLFF	105	VVFF-LLFF	-5
WYYY-IYYY	92	YWYY-YIYY	5	WWYY-IIYY	-5
DEEE-EEEE	10	EDEE-EEEE	2	DDEE-EEEE	-3

TABLE S3 Attribution of each amino acid located at each position to AP_{PRD} , of ten groups of peptides with the largest positive or negative difference (*i.e.*, ϵ) between $AP_{PRD,pe}$ and averaged $AP_{PRD,te}$ (predicted ones). The symbol “ $\uparrow$ ” indicates the case that $AP_{PRD,pe}$ is larger than $AP_{PRD,ave}$ [$AP_{PRD,ave} = (AP_{PRD,teX} + AP_{PRD,teY})/2$, with $Y = X+1$ and $X = 1, 3$] of tetrapeptides, while the symbol “ $\downarrow$ ” indicates the opposite case with a smaller $AP_{PRD,pe}$ than $AP_{PRD,ave}$.

		Pep	P1	P2	P3	P4	P5	AP_{PRD}	AP_{SIM}
$\uparrow$	Te1	KPED	0.182	0.032	0.361	0.265	-	1.260	1.403
		KMED	0.199	0.029	0.349	0.254	-	1.305	1.570
		KMEE	0.186	0.026	0.334	0.310	-	1.256	1.464
		KPEE	0.174	0.032	0.339	0.315	-	1.213	1.332
		KPEE	0.174	0.032	0.339	0.315	-	1.213	1.331
		KPDE	0.183	0.033	0.304	0.326	-	1.261	1.357
		KPEE	0.174	0.032	0.339	0.315	-	1.213	1.314
		KMEE	0.186	0.026	0.334	0.310	-	1.256	1.513
		RPEE	0.152	0.040	0.354	0.327	-	1.219	1.299
		KPED	0.182	0.032	0.361	0.265	-	1.260	1.391
	Te2	PEDF	0.012	0.305	0.277	0.267	-	1.306	1.138
		MEDF	0.002	0.309	0.283	0.265	-	1.289	1.174
		MEEF	0.002	0.300	0.314	0.253	-	1.250	1.108
		PEEF	0.010	0.295	0.307	0.254	-	1.258	1.158
		PEEY	0.009	0.301	0.310	0.247	-	1.262	1.139
		PDEF	0.013	0.259	0.321	0.268	-	1.298	1.128
		PEEW	0.006	0.268	0.275	0.325	-	1.409	1.324
		MEEW	0.003	0.272	0.275	0.324	-	1.389	1.312
		PEEY	0.009	0.301	0.310	0.247	-	1.262	1.124
		PEDY	0.010	0.306	0.277	0.262	-	1.311	1.158
	Pe1	KPEDF	0.138	0.012	0.225	0.177	0.229	1.850	2.007
		KMEDF	0.141	0.015	0.223	0.174	0.220	1.861	1.963
		KMEEF	0.137	0.012	0.211	0.205	0.222	1.814	1.969
		KPEEF	0.135	0.009	0.215	0.209	0.227	1.796	1.899
		KPEEY	0.140	0.007	0.219	0.212	0.223	1.793	1.864
		KPDEF	0.141	0.012	0.196	0.212	0.225	1.832	1.892
		KPEEW	0.117	0.008	0.202	0.191	0.274	1.864	2.051
		KMEEW	0.117	0.015	0.195	0.183	0.268	1.875	2.025
		RPEEY	0.127	0.004	0.221	0.212	0.229	1.792	1.832
		KPEDY	0.137	0.011	0.227	0.176	0.234	1.838	1.890
$\downarrow$	Te3	DPAY	0.357	0.010	0.138	0.327	-	1.580	1.392
		DPAF	0.350	0.013	0.130	0.328	-	1.617	1.400
		DPAY	0.357	0.010	0.138	0.327	-	1.580	1.350
		DVAY	0.333	0.040	0.133	0.298	-	1.651	1.406
		DCAY	0.331	0.023	0.141	0.304	-	1.630	1.340

		EVAY	0.360	0.039	0.137	0.289	-	1.572	1.366
		EPAY	0.391	0.010	0.138	0.314	-	1.470	1.388
		DIAY	0.322	0.047	0.135	0.302	-	1.675	1.407
		DPAF	0.350	0.013	0.130	0.328	-	1.617	1.387
		DMAY	0.340	0.006	0.149	0.318	-	1.555	1.227
	Te4	PAYN	0.039	0.134	0.372	0.172	-	1.914	2.001
		PAFN	0.034	0.135	0.369	0.171	-	1.921	1.892
		PAYQ	0.040	0.132	0.355	0.217	-	1.859	1.991
		VAYQ	0.068	0.129	0.330	0.208	-	1.883	1.862
		CAYQ	0.034	0.134	0.343	0.215	-	1.858	1.826
		VAYQ	0.068	0.129	0.330	0.208	-	1.883	1.849
		PAYN	0.039	0.134	0.372	0.172	-	1.914	1.855
		IAYQ	0.064	0.127	0.338	0.210	-	1.878	1.889
		PAFE	0.023	0.118	0.329	0.326	-	1.724	1.977
		MAYQ	0.011	0.144	0.360	0.229	-	1.813	1.773
	Pe2	DPAYN	0.284	0.004	0.132	0.276	0.144	1.327	1.494
		DPAFN	0.291	0.008	0.131	0.267	0.144	1.354	1.518
		DPAYQ	0.289	0.011	0.127	0.267	0.169	1.310	1.539
		DVAYQ	0.275	0.028	0.129	0.252	0.160	1.368	1.486
		DCAYQ	0.281	0.020	0.132	0.255	0.166	1.354	1.553
		EVAYQ	0.295	0.028	0.130	0.243	0.160	1.340	1.563
		EPAYN	0.306	0.001	0.133	0.272	0.141	1.308	1.474
		DIAYQ	0.267	0.044	0.129	0.245	0.163	1.394	1.476
		DPAFE	0.277	0.003	0.120	0.231	0.227	1.288	1.481
		DMAYQ	0.286	0.000	0.139	0.261	0.170	1.302	1.406

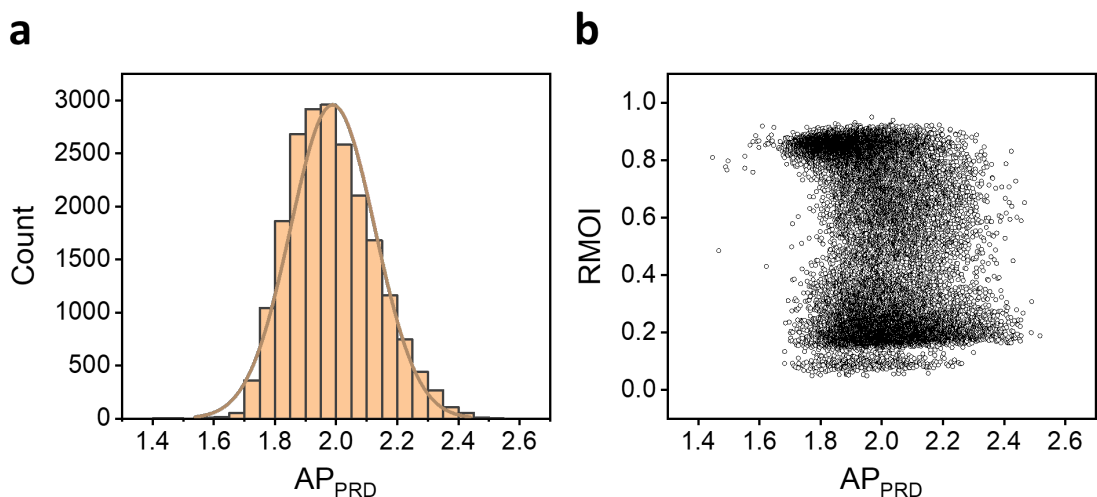


FIGURE S6 a Distribution of predicted AP values ($AP_{PRD} > 1.4$, and 93% of them are greater than 1.8). **b** Correlation between AP_{PRD} and computational RMOI values obtained from 1.25 μ S-CGMD simulations.

Additional discussion regarding the distribution of morphologies:

In this study, we observed a scarcity of peptides with intermediate RMOI values around 0.4, which is unexpected given the known propensity of peptides to form either fiber-like (RMOI ≈ 0.2) or spherical (RMOI ≈ 0.8) structures. Herein, we present a series of analyses aimed at understanding this phenomenon.

1. Sampling Bias Analysis:

Our dataset of 21,050 peptides, generated with an aggregation propensity (AP) threshold of 1.4, may exhibit sampling biases that contribute to the observed scarcity in the intermediate RMOI range. To address this, we propose employing a design of experiment approach to achieve a uniform sampling of peptide sequences, thereby ensuring a more representative distribution.

2. Energy Barrier Considerations:

The intermediate RMOI values (0.4 to 0.5), indicative of complex structures such as net-like configurations, could be influenced by high energy barriers not yet surmounted in our CGMD simulations. It is conceivable that the simulations have not yet reached an equilibrium state between entropy and enthalpy, potentially due to constraints in the spatial or temporal scale of the simulations.

3. Force Field Limitations:

We acknowledge the limitations of the Martini force field in capturing the subtleties of hydrogen bonding and electronic polarizability, which are crucial for an accurate depiction of peptide morphologies. This limitation may be a significant factor contributing to the observed RMOI distribution and warrants further investigation.

4. Literature Corroboration:

However, our findings are supported by existing literature², which predominantly reports peptide assemblies in fiber or spherical forms. Moreover, all-atom molecular dynamics simulations have unveiled complex morphologies with multiple protrusions, as exemplified in Figure S28 of the Supplementary Information of Reference 2, corroborating our third point.

In summary, the scarcity of intermediate RMOI values may be attributed to a combination of factors, including sampling biases, energy barriers, and limitations in the force field model. Future research should consider these aspects to refine the understanding of peptide assembly dynamics. Despite the limitations, our study provides a comprehensive analysis of the RMOI distribution in peptide assemblies, highlighting the need for a more nuanced approach to sampling and simulation. The insights gained from this work contribute to the broader understanding of protein folding and aggregation mechanisms.

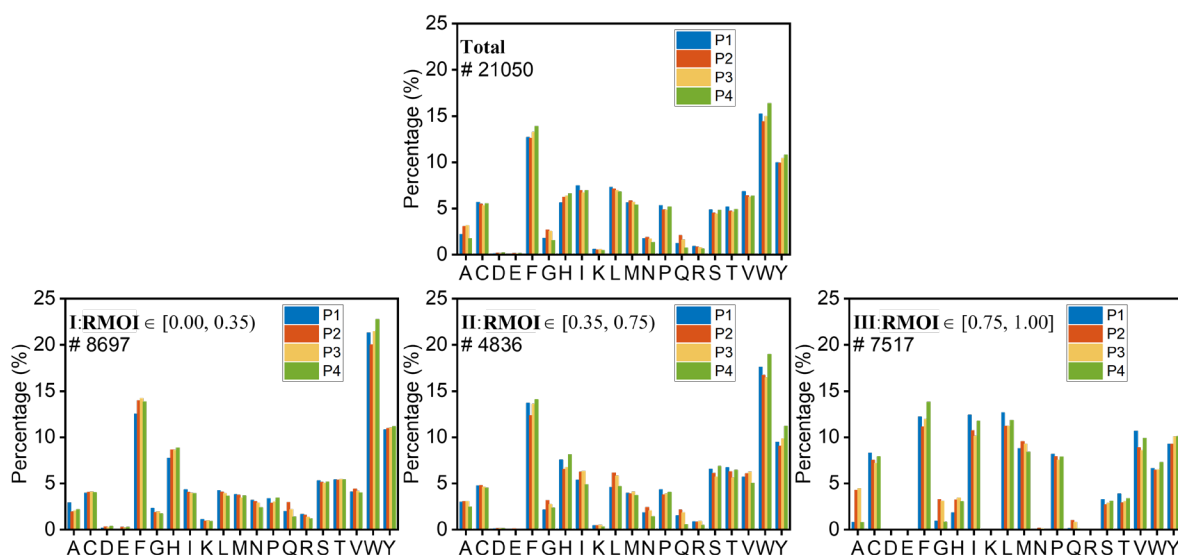


FIGURE S7 The percentage of each amino acid at each position of all 21,050 tetrapeptides (top one figure), as well as the percentage of each amino acid at each position within each range of RMOI values (below three figures).

Notes: The percentage of each amino acid at each position within each range of RMOI will be affected by the original percentage within the total pool of 21,050 peptides. To remove the effect of original percentage difference of each amino acids, in the main text Figure 6e-g, we calculate the *ratio of the number of each amino acid at each position within each RMOI range to the total number of the **specific** amino acid at the position across all ranges (i.e., 21,050 tetrapeptides)*. Namely, we first calculate the number of each amino acid at each position within each range, and then divide the above number by the total number of each amino acid at each position.

Supplementary discussion on correlation between hydrophobicity (logP) and aggregation propensity (AP)

The hydrophobicity is calculated as the sum of the Wimely-White whole-residue hydrophobicity using the equations of $\log P = \sum_{i=1}^n \Delta G_{woc,t,i}$ ($n = 4$ for tetrapeptides, and $n = 5$ for pentapeptides)³, $\Delta G_{woc,t,i}$ (in kcal/mol)⁴ represents the free energy of transfer from water to *n*-octanol for the *i*-th residue.

Figure S8 illustrates a general trend where an increase in hydrophobicity is associated with a corresponding elevation in aggregation propensity (AP) values. However, it is noteworthy that identical AP values are observed across a broad spectrum of hydrophobicity levels, indicating that the relationship is not strictly proportional. In summary, while hydrophobicity may exert a minor positive influence on peptide aggregation, it is not the sole determinant. The aggregation of peptides is a complex phenomenon influenced by a multitude of extrinsic and intrinsic factors, which suggests that hydrophobicity alone cannot fully account for the observed aggregation behavior.

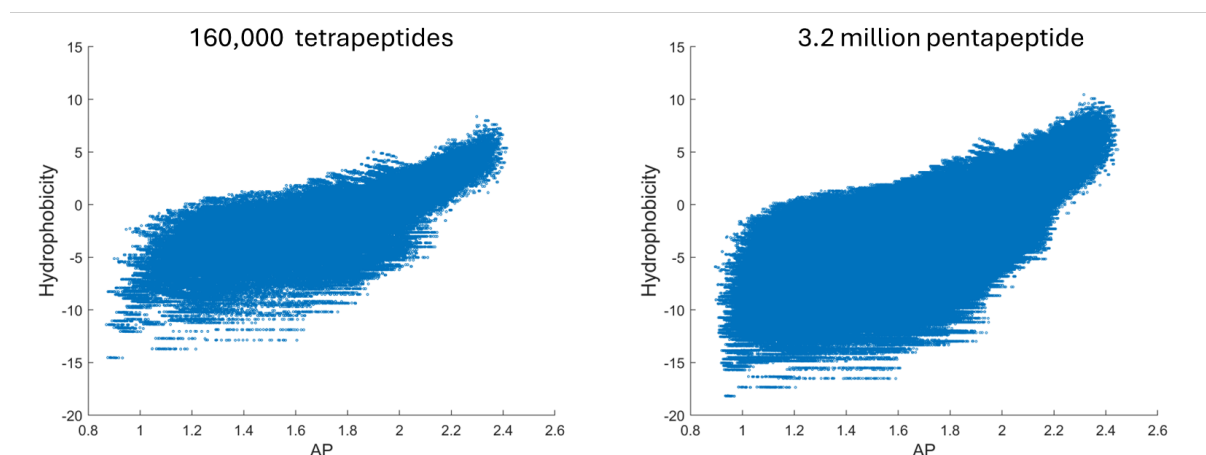


FIGURE S8 Correlation between hydrophobicity and AP, of 160,000 tetrapeptides and 3.2 million pentapeptides.

Supplementary discussion on aggregating morphologies

Standard deviation of computational morphology (RMOI). It should be noted that due to the limited box size, simulation duration, and stochasticity (*e.g.*, initial velocity distribution), a single MD simulation is incapable of producing the most stable and reliable morphology. Therefore, we further carry out parallel simulations to obtain a statistical measure of the accuracy of the obtained morphologies. Among the 21,050 tetrapeptides, we perform parallel simulations on 800 types of randomly selected tetrapeptides for 5 to 10 times with a duration of 1.25 μ s to evaluate the standard deviation (STD) of the RMOI values (Figure S9a). Based on the RMOI and STD values, we classify the peptides into four groups: **Group 1** RMOI $\in$ [0.00, 0.35) and STD $\in$ [0.00, 0.13]; **Group 2** RMOI $\in$ [0.35, 0.75) and STD $\in$ [0.25, 0.40]; **Group 3** RMOI $\in$ [0.75, 1.00) and STD $\in$ [0.00, 0.10], and **Group 4** RMOI $\in$ [0.35, 0.75) and STD $\in$ [0.13, 0.25] (Figure S9b-e). The ratio of each amino acids at each position in each group with respect to the total number of the **specific** amino acid in each position is shown in Figure S9b-e, while the percentage of each amino acid in each group with respect to the **total** number of amino acids in each position is shown in Figure S10.

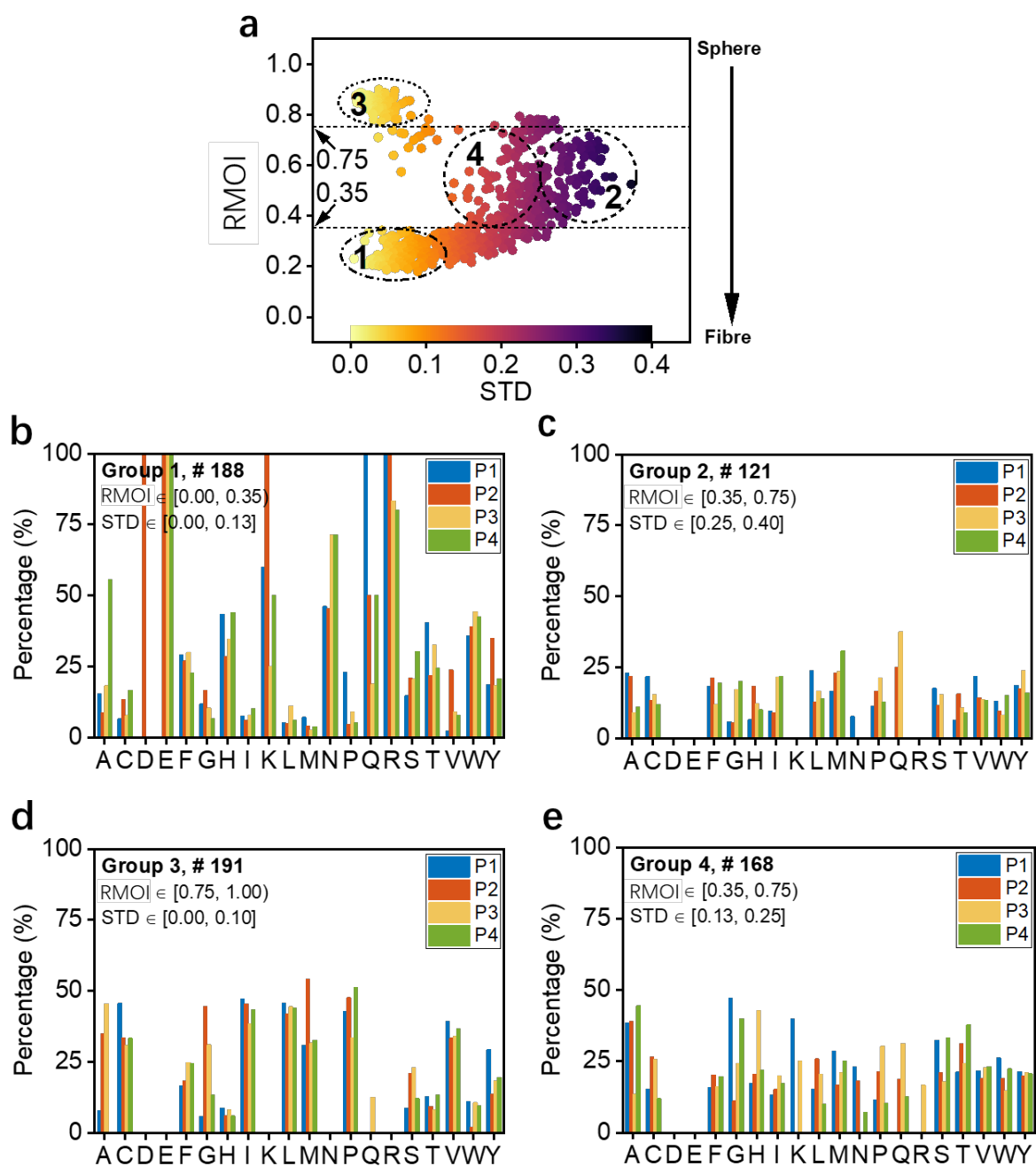


FIGURE S9 Distribution of standard deviation of RMOI (STD) with respect to RMOI values. **b-e** Positional percentage of each amino acid within each RMOI and STD range, with the same calculation method as of Figure 7e-g in the main text (also elaborated in the *Notes* of Figure S7 of this document).

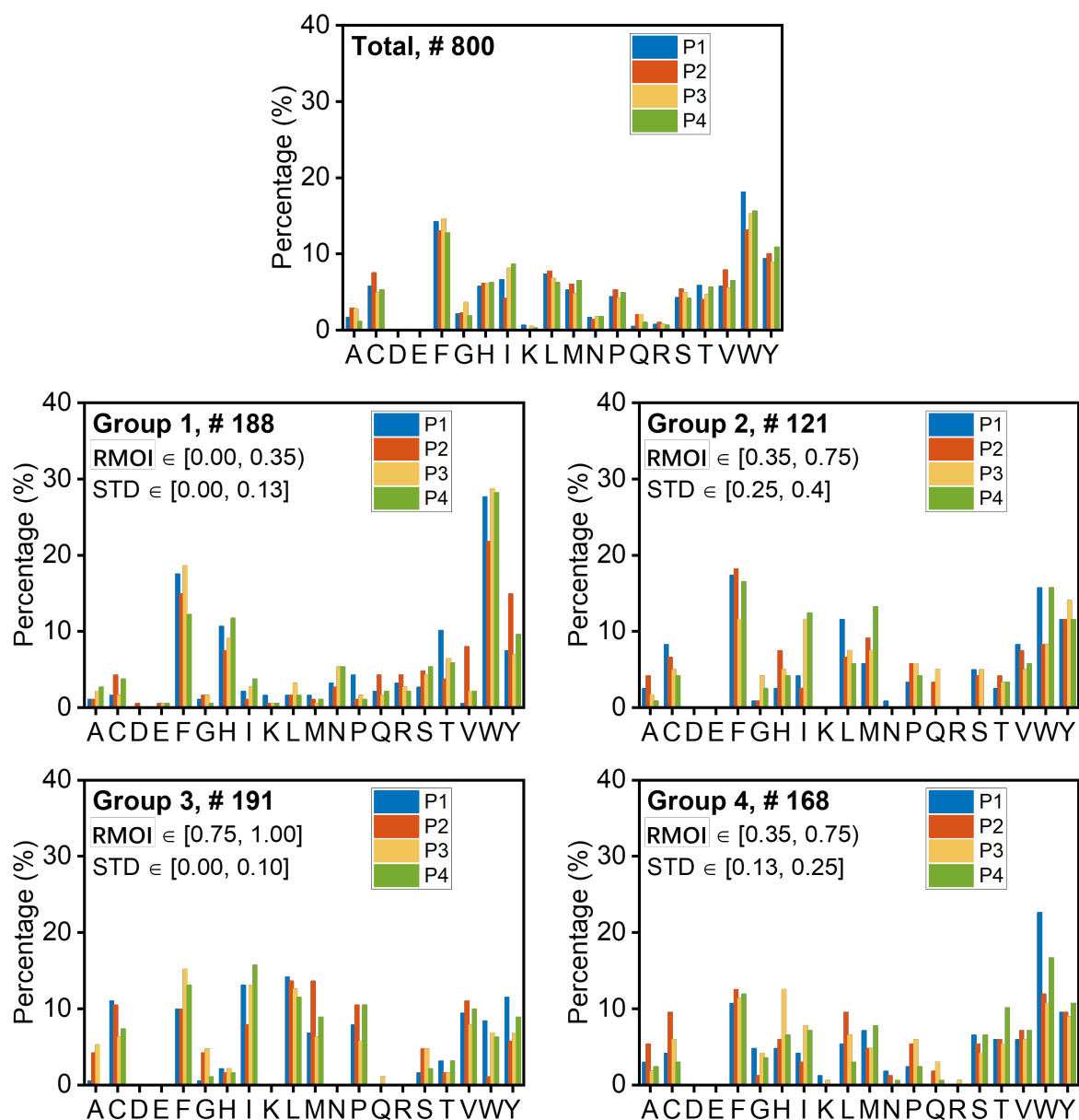


FIGURE S10 The percentage of each amino acid at each position of randomly selected 800 tetrapeptides, as well as the percentage of each amino acid at each position within each range of RMOI value. The STD is calculated from 5 to 10 parallel simulations.

Regarding the determinacy of the morphologies obtained in Figure 7d (of the main text), the maximum STD of the RMOI values obtained in different simulations for a peptide is ~ 0.37 , and most of the peptide morphologies have a STD less than 0.3 (Figure S9a). Peptides with F and W at P3 tend to form robust fiber or sphere structures, those with Y amino acid at P2 tend to form fiber structures, and those with Y at P1 tend to form spheres, with all three cases having a small STD. In contrast, amino acids A, N, and S at the N-termini, G, I, M, and W at the C-termini, H, T, and V at P2, as well as P and Y at P3 can all contribute to a large STD. The large STD can be induced by various interaction types and strengths (*e.g.*, strong Lennard-Jones interaction is more likely result in sphere structures while hydrogen bonding may result in ordered structures), and by hydrophobicity (*e.g.*, strong hydrophobicity may result in amorphous precipitates), *etc.* Therefore, simulations to obtain morphologies with all those amino acids at the specified positions within the sequences should be treated more carefully by performing parallel runs with more testing conditions (*i.e.*, different box size, duration, water model, *etc.*).

Effect of the initial setting of secondary structure on aggregation. In addition to the sequence itself affecting the final morphology, the initial conformational settings of the peptides could also exert an effect since the secondary structures are fixed during aggregation within the Martini force field, so that the transformation of the secondary structure (usually transforming to the β -sheet conformation) for promoting aggregation cannot be embodied. Therefore, we compare the RMOI values of randomly selected 66 peptides among the 800 peptide sequences, averaged over 10 parallel CGMD simulations with different initial

conformational settings (*i.e.*, β -sheet or c-coil, Figure S11a and S11b). The CGMD-generated RMOI and APs with different conformations, as well as the APs predicted by the TRN model, are listed in Supplementary Materials-3. It is generally observed that the peptides with the β -sheet conformation have larger AP values, which is consistent with the reported literature that the peptides are prone to aggregate with the β -sheet conformation⁵. In particular, peptides IMVV, MMMM and IIIR (in the red circle, Figure S11b) cannot even aggregate with the random coil (*i.e.*, c-coil) conformation, and peptides VVVV, PVII, LVFG, GVLF, DVFW, IQII and NTIF (in the blue circle, Figure S11b) show a significantly reduced aggregation tendency. Consequently, the morphologies of the above-mentioned peptides cannot be accurately characterized with the random coil conformation, resulting in large morphology discrepancies between β -sheet and c-coil (Figure S11c), and occasionally a small difference if the AP difference is relatively small (*e.g.*, IQII).

Regarding peptides CTYF, WTIF, PVFW, WALW, WLVW, and WAFY (in the yellow circle, Figure S11b), although they have similar aggregation tendency with different conformations, the generated RMOI have a discrepancy larger than 0.2 (Figure S11c). These peptides (except CTYF) with c-coil conformation tend to form a more spherical structure while forming more fibrous structures with β -sheet conformation. This difference can possibly be attributed to the difference of the properties of the backbone termini (the bead representations are shown in Table S4 in Supplementary Materials-1). With β -sheet conformation, the backbones of above peptides are represented by Nda beads, which have both hydrogen donating and accepting capabilities, whereas with the c-coil conformation, the backbones are mostly represented by P5

bead, which is a polar bead with the strongest Lennard-Jones interaction strength. Therefore, the directional hydrogen bonding plays a dominant role in the formation of the fibrous structures in β -sheet peptides, resulting in a smaller RMOI value than those with c-coil conformation.

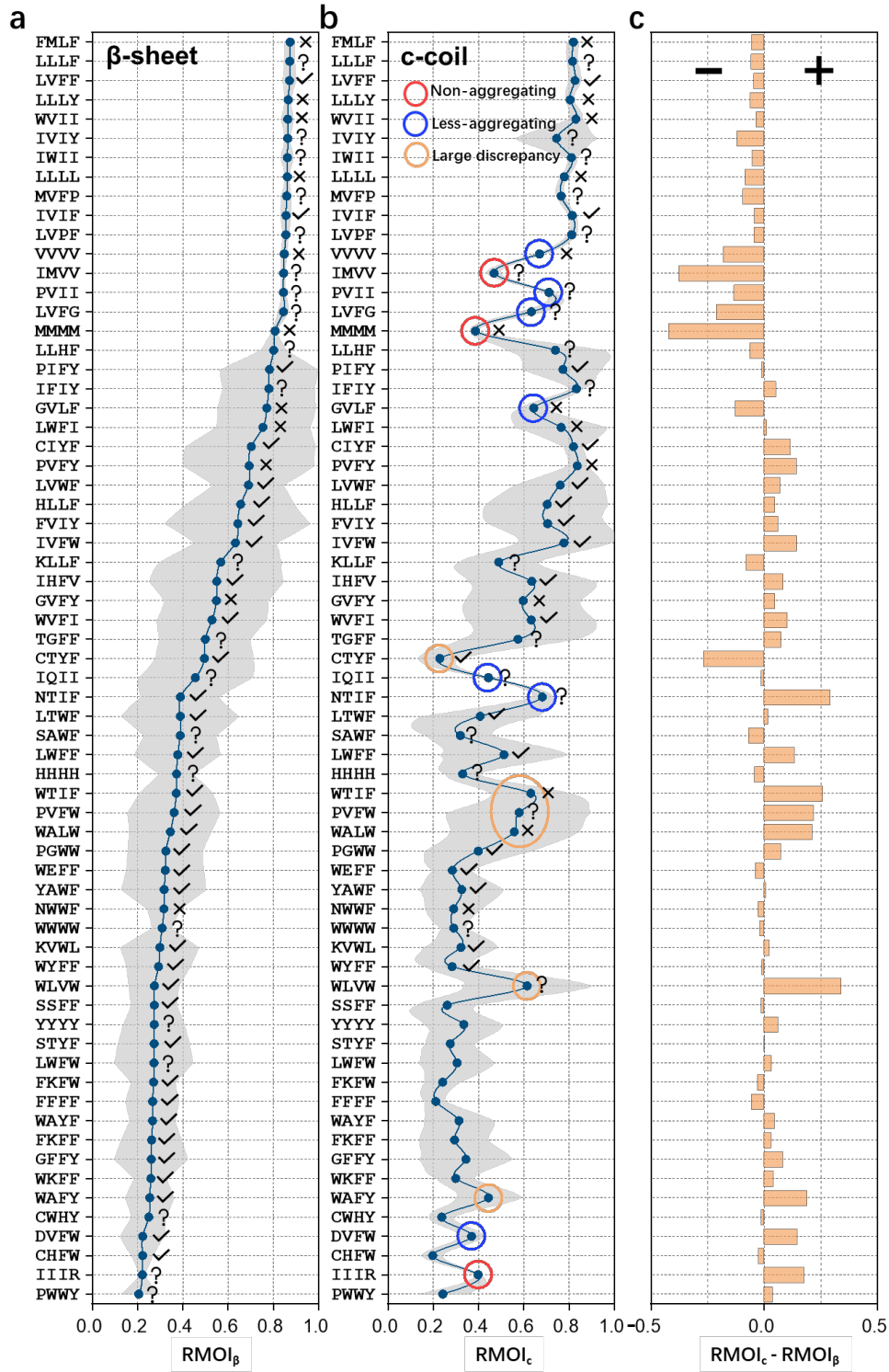


FIGURE S11 a-b RMOI values of the largest cluster of aggregated peptides with β -sheet and c-coil conformation. The grey shadow indicates the uncertainties of the RMOI values calculated by 10 parallel simulations. **c** Difference of RMOI value of the same type of peptide between different conformations ($= \text{RMOI}_c - \text{RMOI}_{\beta}$). Here the RMOI is the mean value over ten parallel simulations.

Comparisons between simulations and experiments. More importantly, we would like to provide an assessment of the discrepancy between the computational and experimental morphologies at a statistical level, by comparing the TEM images (Supplementary Materials-3, obtained from our previous work¹) and the morphology images generated in this work. We use the symbol ✓ to denote a match between simulation and experiment by visual inspection, the symbol ✕ to denote a mismatch, and ? to denote an unidentified relationship due to the ambiguity of the experimental morphology (Figure S11b). Among the 66 peptides with β -sheet conformation (/c-coil conformation), 11 peptides (/14) do not match, 21 (/25) peptides are unidentified or unambiguous of comparison, while 34 (/27) morphologies of peptides generally match each other, resulting an approximately 51% (/41%) of the prediction accuracy of morphologies between simulation and experiment. Assigning β -sheet conformation to peptides achieves a slightly higher accuracy. The mismatched peptides generally have RMOI values greater than 0.5 (~ a close-to-spherical structure), while the experimental data indicate fiber-like structures.

The discrepancy between the morphologies formed in simulations and experiments can be attributed to three main reasons. The first is the temporal and spatial limitation of the CGMD simulation. The short peptides self-assemble in two steps in the real world, *i.e.*, oligomerization into clusters and subsequent self-assembly into supramolecules⁶, while the morphologies generated by the simulations may still be in the first step of oligomerization within the 1.25 μ s of simulations. This may explain why the close-to-spherical structures formed in simulations do not match the experimental results. These close-to-spherical structures may further self-

assemble into fibers or tubes. In addition, the simulation size is typically on the order of several nanometers, which is about two orders of magnitude smaller than that in experiments, thus there are not sufficient molecules to self-assemble into real-world supramolecular structures. In summary, much extended simulation time and size should be a necessity to obtain accurate structural information⁷⁻⁹.

The second cause could be the limitation of the Martini force field. The Martini 2.2 force field used in our simulations lacks an accurate description of hydrogen bonding and electronic polarizability, both of which are important interactions for mediating self-assembly morphologies¹⁰⁻¹². These interactions have been fine-tuned in Martini 3 and it has been successfully applied to reproduce the preferential configuration of amide-linked tetramers (*i.e.*, alternating stacks of 1,5-dialkoxynaphthalene and naphthalene diimide¹³). However, according to a recent study¹⁴, Martini 3 results in smaller AP values of short peptides and a more disordered and hydrophobically-driven aggregation of diphenylalanine (*i.e.*, FF), while the Martini 2.2 outperforms all other versions of the Martini force field family in simulating the self-assembly of the short peptides (specifically, di- and tripeptides) in aqueous environments. Further testing of the Martini force field can be performed on the longer peptides containing more than three amino acids.

It should also be mentioned that the morphologies of self-assembled peptides in experiments are affected by various factors such as temperature, pH, solvent type, standing time, *etc.* Regarding the stand time, the stable morphologies could be achieved after several hours, days,

or even months of standing time, while the morphologies reported here may still be intermediate structural states, which may also be one of the possible causes of the RMOI discrepancy between simulation and experiment.

TABLE S4 Bead representation of the peptides CTYF, WTIF, PVFW, WALW, WLVW, and WAFY with β -sheet and c-coil conformation, generated by Martini force field version 2.2.

Peptide	Amino acid	Bead type	β -sheet	c-coil
CYS THR TYR PHE (CTYF)	CYS	BB	Nda	P5
	CYS	SC1	C5	C5
	THR	BB	Nda	P5
	THR	SC1	P1	P1
	TYR	BB	Nda	P5
	TYR	SC1	SC4	SC4
	TYR	SC2	SC4	SC4
	TYR	SC3	SP1	SP1
	PHE	BB	Nda	P5
	PHE	SC1	SC5	SC5
	PHE	SC2	SC5	SC5
	PHE	SC3	SC5	SC5
TRP THR ILE PHE (WTIF)	TRP	BB	Nda	P5
	TRP	SC1	SC4	SC4
	TRP	SC2	SNd	SNd
	TRP	SC3	SC5	SC5
	TRP	SC4	SC5	SC5
	THR	BB	Nda	P5
	THR	SC1	P1	P1
	ILE	BB	Nda	P5
	ILE	SC1	AC1	AC1
	PHE	BB	Nda	P5
	PHE	SC1	SC5	SC5
	PHE	SC2	SC5	SC5
	PHE	SC3	SC5	SC5
PRO VAL PHE TRP (PVFW)	PRO	BB	N0	P4
	PRO	SC1	C3	C3
	VAL	BB	Nda	P5
	VAL	SC1	AC2	AC2
	PHE	BB	Nda	P5
	PHE	SC1	SC5	SC5
	PHE	SC2	SC5	SC5
	PHE	SC3	SC5	SC5
	TRP	BB	Nda	P5
	TRP	SC1	SC4	SC4
	TRP	SC2	SNd	SNd
	TRP	SC3	SC5	SC5
	TRP	SC4	SC5	SC5
TRP	TRP	BB	Nda	P5

 ALA LEU TRP (WALW)	TRP	SC1	SC4	SC4
	TRP	SC2	SNd	SNd
	TRP	SC3	SC5	SC5
	TRP	SC4	SC5	SC5
	ALA	BB	N0	P4
	LEU	BB	Nda	P5
	LEU	SC1	AC1	AC1
	TRP	BB	Nda	P5
	TRP	SC1	SC4	SC4
	TRP	SC2	SNd	SNd
	TRP	SC3	SC5	SC5
	TRP	SC4	SC5	SC5
TRP LEU VAL TRP (WLVW)	TRP	BB	Nda	P5
	TRP	SC1	SC4	SC4
	TRP	SC2	SNd	SNd
	TRP	SC3	SC5	SC5
	TRP	SC4	SC5	SC5
	LEU	BB	Nda	P5
	LEU	SC1	AC1	AC1
	VAL	BB	Nda	P5
	VAL	SC1	AC2	AC2
	TRP	BB	Nda	P5
	TRP	SC1	SC4	SC4
	TRP	SC2	SNd	SNd
	TRP	SC3	SC5	SC5
	TRP	SC4	SC5	SC5
TRP ALA PHE TYR (WAFY)	TRP	BB	Nda	P5
	TRP	SC1	SC4	SC4
	TRP	SC2	SNd	SNd
	TRP	SC3	SC5	SC5
	TRP	SC4	SC5	SC5
	ALA	BB	N0	P4
	PHE	BB	Nda	P5
	PHE	SC1	SC5	SC5
	PHE	SC2	SC5	SC5
	PHE	SC3	SC5	SC5
	TYR	BB	Nda	P5
	TYR	SC1	SC4	SC4
	TYR	SC2	SC4	SC4
	TYR	SC3	SP1	SP1

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